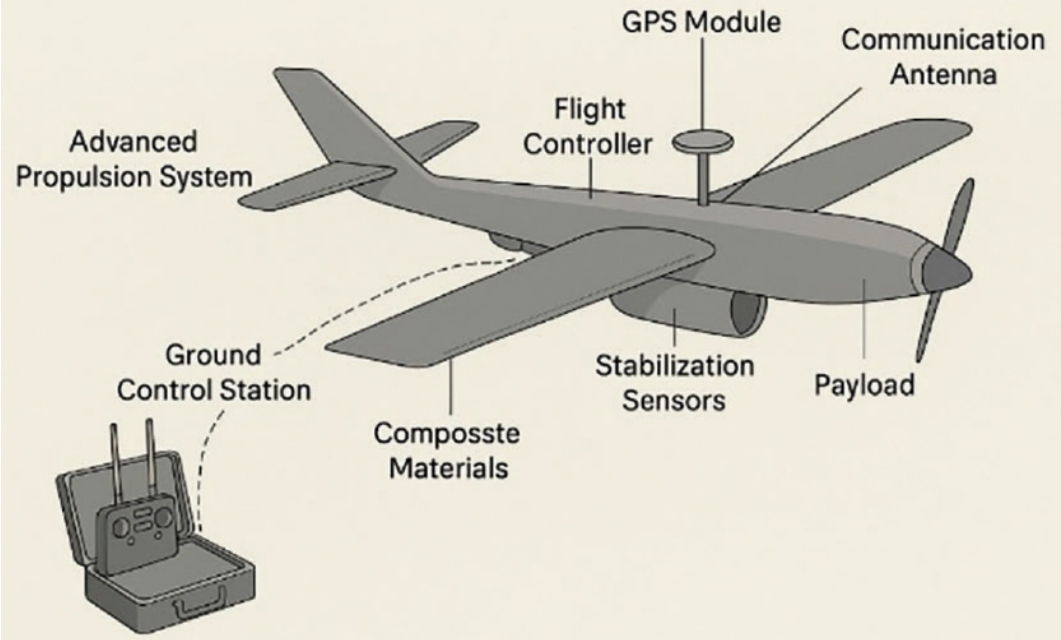


Figure 3.2 Integrated architecture of second-generation UAV systems.⏎

3.4 Applications of Second-generation UAV Systems

The second generation of unmanned aerial vehicle (UAV) systems marked a transformative period in the evolution of drone technology, expanding their use from specialized military operations to a wide range of civilian, scientific, and industrial applications. Between the 1950s and early 2000s, the combination of improved propulsion systems, digital communication, GPS navigation, and real-time telemetry allowed drones to become more reliable, versatile, and mission-capable. These advancements enabled UAVs to play an increasingly vital role in reconnaissance, surveillance, data acquisition, and environmental monitoring, establishing their importance in both strategic defense and non-military sectors.

3.4.1 Military and defense applications

- *Reconnaissance and surveillance*: Second-generation drones were extensively deployed for battlefield reconnaissance and intelligence